

C Problem Set

C Design Problems

1. Reverse a String

- File: `reverse\_string`
- Description: Swap chars in place

2. Find Max in Array

- File: `max\_in\_array`
- Description: Linear scan $O(n)$

3. Matrix Multiplication

- File: `matrix\_mul`
- Description: Nested loops $N \times N$

4. Count Bits

- File: `bit\_count`
- Description: Brian Kernighan's algorithm

5. Linked List

- File: `linked\_list`
- Description: Insert/delete nodes

6. Stack with Array

- File: `stack\_array`
- Description: Push/pop/peek

7. Queue with Two Stacks

- File: `queue\_stacks`
- Description: Enqueue/dequeue

8. Implement atoi()

- File: `atoi\_impl`
- Description: String to integer

9. Binary Search

- File: `binary\_search`
- Description: Iterative/recursive

10. Memory Copy with Overlap

- File: `memmove\_impl`
- Description: Implement memmove